

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element H Initial Template

Element H: Prototype Testing and Data Collection Plan

Testing Details (H1, H2, H3, H4)

	requirement	testing setup and procedure	test cases and number of trials	expected range of results	pass/fail criteria, goal
1.	Height Must be over 12 feet long.	First, have a tape measurer and the prototype. Next, put the tape measurer on the top end of the stick and run the tape measure all the way down to the bottom. Finally, record the height of the stick from the tape measurer.	1	12-15 feet	Pass: The height is in the expected range of 12-15 feet in height or is above the range. Fail: The height is under the expected range of 12-15 feet. Goal: Have the prototype to be in the range of 12-15 feet as that should be enough for the prototype to work. We also have an extension piece which could be used to reach the height.
2.	Stability Must be structurally straight and stable.	First, have the stick prototype held in an open space horizontally and parallel to the ground. Next, record if there is any bends or weak spots in the design. Closely examine where the segments connect as there could be weak spots.	1	The stick bends at multiple locations and is not stable. The stick bends at few locations and is mostly stable. The stick is	Pass: The stick is either perfectly straight or has few spots of weakness or bends. Fail: The prototype has several bends

		Finally, record if the stick prototype has a bend angle or not.		consistently straight and is secure.	and isn't stable. Goal: To have our prototype be straight in shape as it will be the most stable when held vertically.
3.	Mobility/Transportation The prototype must be transported 20 feet and reassembled with in 2 minutes.	First, have the prototype fully together. Next, disassemble the prototype into the 4 segments. Take the segments and move them to their final destination. Finally, put the prototype can back together.	1	The stick can be transported successfully. There are multiple problems involving transportation like weight and distance. The prototype won't come apart. The prototype won't come back together. We complete the task but in over 2 minutes.	Pass: The prototype can be disassembled, transported 20 feet, and constructed successfully or with minimal errors with in the time frame. Fail: The prototype can't be transported, put back together, is not secure, or can't be disassembled within the time frame. Goal: To have the prototype be transported from A to B in a timely manner without problems. This is a key component of our design and it is important for the stakeholders.

Feedback From Field Experts (H5)

Guidance to students about this section of Element H.

- a) Do not incorporate their feedback in the table above. Rather, make a copy of the table in the section above, paste it into the section below, and make all revisions only in the table below. This provides evidence that you incorporated the feedback and shows your testing plan's progression.

Revised Testing Details That Incorporate Field Expert Feedback (H5)

- **Mrs. Armstrong** is a mechanical engineer who works with design principles and physics every day. She recommended that we time our data for the transportation segment.
- **Mr. Butchko** is a civil engineer and uses engineering principles every day to design buildings. He told us that we should find weak points in the stability at where the different parts meet.

Changes to the table are in **Bold**.

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	5	4	3	2	1	0

1. Addressing Requirements. The final prototype Testing Plan addresses _____ of the <i>high priority</i> design requirements.	<i>all or nearly all</i>	<i>many</i>	<i>some</i>	<i>a few</i>	<i>one</i>	<i>none - or is missing altogether</i>
2. Conduct of Tests. The Testing Plan _____ in describing the conduct of tests (through physical and/or mathematical modeling).	<i>is effective</i>	<i>is generally effective</i>	<i>is only adequate</i>	<i>is partially effective</i>	<i>is at least minimally effective</i>	<i>is void</i>
3. Quantity of Tests Planned. The Testing Plan defines _____ of <i>all</i> design requirements based on the instructional context.	sufficient quantity to feasibly test 100%	sufficient quantity to feasibly test 80%	sufficient quantity to feasibly test 60%	sufficient quantity to feasibly test 40%	<i>sufficient quantity to feasibly test 20%</i>	<i>no feasible test</i>
4. Logic of the Plan. The Testing Plan provides _____ .	<i>a logical and well-developed explanation</i>	<i>a logical and generally developed explanation</i>	<i>a generally logical and adequately developed explanation</i>	<i>a somewhat logical and partially-developed explanation</i>	<i>at least a generally logical and/or partially-developed explanation</i>	<i>no explanation</i>
5. Use of Field Experts. The Testing Plan is confirmed by _____.	<i>more than two field experts with feedback provided and incorporated</i>	<i>two field experts with feedback provided and incorporated</i>	<i>one field expert with feedback provided and incorporated</i>	<i>one field expert with feedback provided</i>	<i>one field expert but no response was documented</i>	<i>Field expertise is not mentioned or addressed and field experts are not consulted.</i>